Experiment - 2

Aim :- To convert the ER diagram of an Indian e-commerce platform into a relational schema and implement it in MySQL using primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

Source Code :-

1.) Create Database

```
CREATE DATABASE ecommerce_db ;
USE ecommerce_db ;
```

2.) Create Customer Table

```
CREATE TABLE Customer (
Customer_ID INT PRIMARY KEY,
First_Name VARCHAR(30) NOT NULL,
Middle_Name VARCHAR(30),
Last_Name VARCHAR(30) NOT NULL,
Email VARCHAR(30) NOT NULL,
Date_of_Birth DATE
);
```

3.) Create Customer_Phone Table

Since Phone is a multi-valued attribute, it is converted into a separate table.

```
CREATE TABLE Customer_Phone (
Customer_ID INT,
Phone VARCHAR(15),
PRIMARY KEY (Customer_ID, Phone),
FOREIGN KEY (Customer_ID)
```

```
    REFERENCES Customer (Customer_ID)
    ON DELETE CASCADE
);
```

4.) Create Address Table

```
CREATE TABLE Address (
    Address_ID INT PRIMARY KEY,
    Customer_ID INT NOT NULL,
    House_No VARCHAR(20),
    Street VARCHAR(100),
    City VARCHAR(50) NOT NULL,
    State VARCHAR(50) NOT NULL,
    Pincode VARCHAR(50) NOT NULL,
    Address_Type VARCHAR(20),
    FOREIGN KEY (Customer_ID)
        REFERENCES Customer (Customer_ID)
        ON DELETE CASCADE
);
```

5.) Create Seller Table

```
CREATE TABLE Seller (
    Seller_ID INT PRIMARY KEY,
    Seller_Name VARCHAR(100) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Phone VARCHAR(15),
    GSTIN VARCHAR(20) UNIQUE,
    Rating DECIMAL(2,1)
);
```

6.) Create Category Table

```
CREATE TABLE Category (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50) NOT NULL UNIQUE,
    Description VARCHAR(200)
);
```

7.) Create Product Table

```
CREATE TABLE Product (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR (100) NOT NULL,
    Price DECIMAL (10, 2) NOT NULL,
    Stock INT NOT NULL,
    Description VARCHAR (200),
    Product_Type VARCHAR (30),
    Seller_ID INT NOT NULL,
    Category_ID INT,
    FOREIGN KEY (Seller_ID)
      REFERENCES Seller (Seller_ID)
      ON DELETE CASCADE,
    FOREIGN KEY (Category_ID)
      REFERENCES Category (Category_ID)
      ON DELETE SET NULL
);
```

8.) Create Electronic_Product Table.

```
CREATE TABLE Electronic_Product (
    Product_ID INT PRIMARY KEY,
    Brand VARCHAR (50),
    Warranty_Period INT,
    FOREIGN KEY (Product_ID)
      REFERENCES Product (Product_ID)
      ON DELETE CASCADE
);
```

9.) Create Clothing_Product Table

```
CREATE TABLE Clothing_Product (
    Product_ID INT PRIMARY KEY,
    Size VARCHAR (10),
    Material VARCHAR (50),
    FOREIGN KEY (Product_ID)
```

```
    REFERENCES Product (Product_ID)
    ON DELETE CASCADE
);
```

10.) Create Orders Table

```
CREATE TABLE Orders (
    Order_ID INT PRIMARY KEY,
    Customer_ID INT NOT NULL,
    Order_Date DATE NOT NULL,
    Order_Status VARCHAR(30) NOT NULL,
    Total_Amount DECIMAL(12,2) NOT NULL,

    FOREIGN KEY (Customer_ID)
      REFERENCES Customer (Customer_ID)
      ON DELETE CASCADE
);
```

11.) Create Order_Item Table

```
CREATE TABLE Order_Item (
    Order_ID INT,
    Product_ID INT,
    Quantity INT NOT NULL,
    Unit_Price DECIMAL(10,2) NOT NULL,
    Discount DECIMAL(5,2) DEFAULT 0,

    PRIMARY KEY (Order_ID, Product_ID),

    FOREIGN KEY (Order_ID)
      REFERENCES Orders (Order_ID)
      ON DELETE CASCADE,
    FOREIGN KEY (Product_ID)
      REFERENCES Product (Product_ID)
      ON DELETE CASCADE
);
```

12.) Create Payment Table

```
CREATE TABLE Payment (
    Payment_ID INT PRIMARY KEY,
    Order_ID INT NOT NULL UNIQUE,
    Payment_Date DATE,
    Amount DECIMAL(12,2) NOT NULL,
    Payment_Method VARCHAR(30),
    Payment_Status VARCHAR(30),
    Transaction_ID VARCHAR(100) UNIQUE,

    FOREIGN KEY (order_ID)
        REFERENCES Orders (Order_ID)
        ON DELETE CASCADE
);
```

13.) Create Delivery Table

```
CREATE TABLE Delivery (
    Delivery_ID INT PRIMARY KEY,
    Order_ID INT NOT NULL UNIQUE,
    Address_ID INT,
    Delivery_Date DATE,
    Delivery_Status VARCHAR(30),
    Tracking_Number VARCHAR(100) UNIQUE,
    Delivery_Partner VARCHAR(50),

    FOREIGN KEY (Order_ID)
        REFERENCES Orders (Order_ID)
        ON DELETE CASCADE,
    FOREIGN KEY (Address_ID)
        REFERENCES Address (Address_ID)
        ON DELETE SET NULL
);
```

Insert SAMPLE DATA

Customer

```
INSERT INTO Customer
VALUES
(1,'Ayush', NULL, 'Tripathi', 'ayush@gmail.com', '2007-05-15'),
(2,'Rahul', NULL, 'Sharma', 'rahul@gmail.com', '2006-06-20'),
(3, 'Priya', NULL, 'Verma', 'priya@gmail.com', '2007-01-10'),
```

Customer Phone

```
INSERT INTO Customer_Phone
VALUES
(1, '9876543610')
(1, '9892534567'),
(2, '9988667766'),
(3, '8081096757');
```

Address

```
INSERT INTO Address
VALUES
(101, 1, '12A', 'MG Road', 'Kanpur', 'UP', '208001', 'Home'),
(102, 1, '45B', 'Civil Lines', 'Kanpur', 'UP', '208001', 'Work'),
(103, 2, '22', 'Main Road', 'Lucknow', 'UP', '226001', 'Home'),
```

Seller

```
INSERT INTO Seller
VALUES
(201, 'Tech World', 'tech@gmail.com', '9876543000', 'GSTIN', 4.5),
(202, 'Fashion Hub', 'fashion@gmail.com', '9876543111', 'GSTIN', 4.2),
```

Category

```
INSERT INTO Category
VALUES
(301, 'Electronics', 'Electronics Products'),
(302, 'Clothing', 'Clothing and fashion products');
```

Product

```
INSERT INTO Product
VALUES
(401, 'Laptop', 65000.00, 20, 'Gaming Laptop', 'Electronics',
201, 301),
(402, 'Smartphone', 25000.00, 50, 'Android Smartphone',
'Electronics', 201, 301),
(403, 'T-Shirt', 799.00, 100, 'Cotton T-Shirt', 'Clothing',
202, 302);
```

Electronic Product

```
INSERT INTO Electronic_Product
VALUES
(401, 'Dell', 24),
(402, 'Samsung', 12);
```

Clothing Product

```
INSERT INTO Clothing_Product
VALUES
(403, 'L', 'Cotton');
```

Orders

```
INSERT INTO Orders
VALUES
(501, 1, '2026-08-18', 'Confirmed', 65000.00),
(502, 2, '2026-08-18', 'Delivered', 25000.00);
```

Order Item

```
INSERT INTO Order_Item
VALUES
(501, 401, 1, 65000.00, 0),
(502, 402, 1, 25000.00, 500);
```

Payment

```
INSERT INTO Payment
VALUES
(601, 501, '2026-08-18', 65000.00, 'UPI', 'Successful',
'TXN1001'),
(602, 502, '2026-08-18', 24500.00, 'Credit Card',
'Successful', 'TXN1002');
```

Delivery.

```
INSERT INTO Delivery
VALUES
(701, 501, 101, NULL, 'Processing', 'TRK1001', 'Delhivery');
(702, 502, 103, '2026-08-20', 'Delivered', 'TRK1002',
'BlueDart');
```

REFERENTIAL INTEGRITY VIOLATIONS

1) foreign key Violation

```
INSERT INTO Orders
VALUES
(503, 999, '2026-08-18', 'Confirmed', 1000.00);
```

2) UNIQUE Constraint Violation

```
INSERT INTO Customer
VALUES
(4, 'Amit', NULL, 'Kumar', 'ayush@gmail.com', '2006-03-12');
```

3) NOT NULL Constraint Violation

```
INSERT INTO Customer
VALUES
(5, NULL, NULL, 'Singh', 'singh@gmail.com', '2006-04-10');
```

Display Customers

Output:

Customer_ID	First_Name	Last_Name	Email	Date_of_Birth
1	Ayush	Tripathi	ayush@gmail.com	2007-05-15
2	Rahul	Sharma	rahul@gmail.com	2006-08-20
3	Priya	Verma	priya@gmail.com	2007-01-10

Display Products with Sellers

Output:

Product_Name	Price	Seller_Name
Laptop	65000.00	Tech World
Smartphone	25000.00	Tech World
T-Shirt	799.00	Fashion Hub

Display Orders with Customer Names

Order_ID	First_Name	Last_Name	Total_Amount	Order_Status
501	Ayush	Tripathi	65000.00	Confirmed
502	Rahul	Sharma	25000.00	Delivered

